

Nightingale

Technical Brief

STATUS	BASELINE	LAST UPDATED
Prototype - verified with limits	HEAD 21a9a4a + submission closeout	03 Sep 2026
SCOPE	Synthetic-data prototype for a shared longitudinal care record; not production medical software.	

1. Abstract

Nightingale is a shared longitudinal care record that turns fragmented dated notes into one verifiable patient story. Real-world Events remain the Timeline unit; parallel Artifacts preserve raw, AI-scribed, clinician, staff, and patient-facing representations without overwriting one another. Glance surfaces current priorities while every derived statement resolves to an exact source Span.

2. Design principles

- Timeline = what happened. Glance = what matters now. Patient View = what the patient needs to know or do.
- Server-side role, clinic, and patient scope is authoritative; the frontend is never the security boundary.
- AI assists language only. Provider output cannot create authority, bypass safety, or become its own source.

3. Proposed architecture

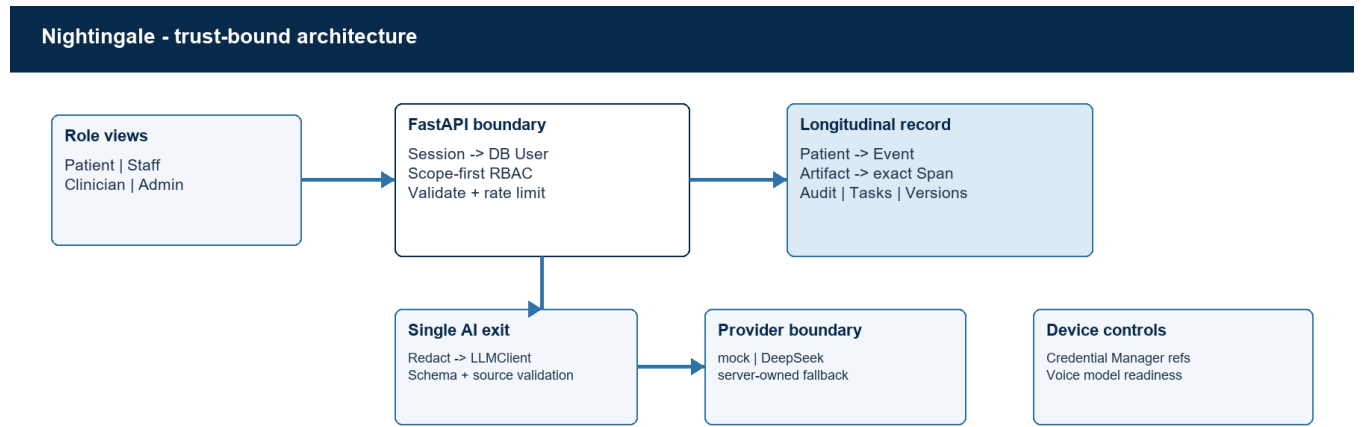


Figure 1. Trust boundaries, durable record, and the single AI egress path.

Authority boundary	The server owns RBAC, safety, state, Task authority, provenance, and limits. AI/patient-derived content never overwrites clinician or staff Artifacts.
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4. Data model and provenance

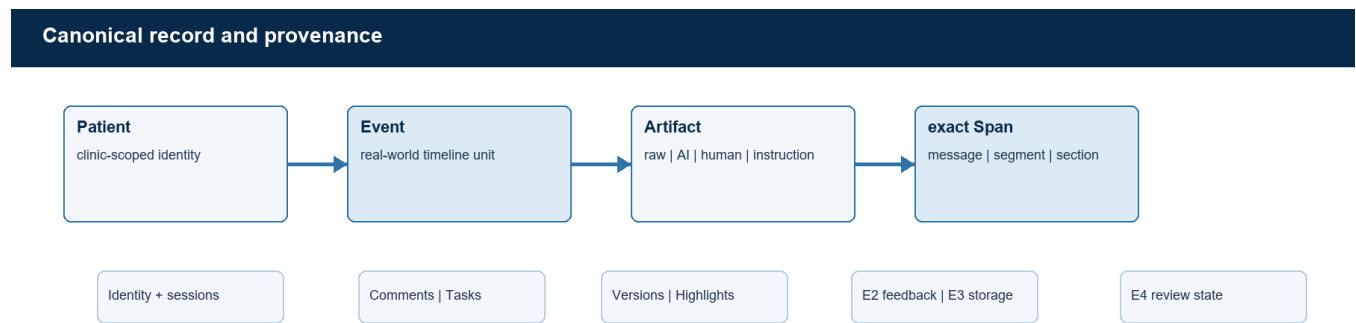


Figure 2. Candidate Brief Entry maps to Event + parallel Artifacts; exact Span is the trust anchor.

5. Role journeys

Role	Primary safe journey
Clinician	Read Glance in under 10 seconds -> inspect exact source -> review Event lifecycle and collaboration.
Staff	Add same-clinic Staff Notes, Nurse Consults, comments, and Tasks without assuming clinician authority.
Patient	Read instructions and Tasks -> complete a bounded raw-first Check-in -> return/correct/confirm safely.

6. RBAC and authority

Role	Can	Cannot
Patient	Own safe view, visible Tasks, own Check-in	Internal notes/comments; other patient; clinical authoring
Staff	Same-clinic staff notes, Nurse Consult, Tasks	Clinician authorship; other clinic
Clinician	Same-clinic review, notes, Copilot	Staff authorship; other clinic
Admin	Identity/session/invite and device settings	Clinical Note, Copilot, Task, or plan authoring

7. Patient Multi-turn Check-in

active	awaiting confirmation	submitted	safety escalated	abandoned
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Every patient message is committed first with a stable message_id. At most four clarification questions are allowed. Deterministic safety and medical-request rules run after raw persistence; confirmation creates the formal AI Summary, and every candidate quote must resolve to one patient message or be dropped.

8. Verification status

Capability	Status	Evidence boundary
E1 Role workspaces	IMPLEMENTED_AND_VERIFIED	Role-specific UI, RBAC, scope, and state isolation.
E2 + SL1-SL3 ranking	IMPLEMENTED_WITH_LIMITS	Formal Glance is deterministic base-only; frozen synthetic Shadow models and the observed-feedback bridge do not serve a model.
E3 Data-decay shadow	IMPLEMENTED_AND_VERIFIED	Protection-first shadow payload; not total DB savings.
E4 Voice + consult review	IMPLEMENTED_WITH_LIMITS	Default-off; three review attestations. Local ASR mechanics passed 2/2; clinical multilingual/noisy accuracy remains unverified.
Patient Multi-turn Check-in	IMPLEMENTED_AND_VERIFIED	Raw-first, bounded safety, concurrency, RBAC, exact provenance.
Official feedback scenarios 1–16	5 SURVIVE / 9 PARTIAL / 2 DO NOT SURVIVE	#1 phone/WhatsApp identity and #7 streaming in-consult alert are not implemented. Exact first breaks and limits are recorded in the readiness ledger.

9. Measured evidence

Signal	Observed	Boundary
Backend	712 passed / 2 skipped	714 collected; default skips require ignored ASR inputs. F Final 9/9 passed.
Frontend	Build passed / 112 modules	TypeScript no-emit and Vite production build passed.
Live Provider + local ASR	DeepSeek 4/4 anchors; ASR 2/2	One synthetic Provider run and one generated-audio mechanics run; not clinical validation.
Glance	Layer B P95 5.115 ms	100 local samples after 10 warm-ups; target <= 300 ms.

10. Provider, ASR, learning, and security boundary

- Mock and fallback journeys are verified. One current synthetic mixed-language DeepSeek run passed strict schema and exact provenance (4/4 quotes). This is not clinical validation or a full live UI journey.
- Frozen D3/D4 evaluations remain Provider-independent; live DeepSeek evidence is recorded separately.
- Local faster-whisper is default-off and has no diarization. Explicit real-ASR tests passed 2/2 with ignored inputs, but two Malay phrases were misrecognized. Doctor Consult requires three source-review attestations. Physical microphone, noise and clinical multilingual validity remain NOT_RUN.
- Formal Glance remains deterministic base-only. SL1 records all candidate decisions, SL2 uses frozen synthetic role models in Shadow, and SL3 has no eligible real-feedback model. Names, IC/ID patterns, and phones are redacted before LLM egress; SQLCipher, TLS, metadata-only audit, and Credential Manager references protect the local prototype.

11. Trade-offs

Decision	Why
Deterministic ranking before LLM ranking	Transparent, fast, reproducible, and auditable under a 72-hour prototype constraint.
Shadow learning controls	Frozen Shadow only; caps, provenance, replay, freeze and rollback. It never changes production Glance.